

electronics within the device but the whole integration is done through software. There are broadcast solutions, on-demand content, a slick user interface, support for a variety of formats and codecs. In the last two years, we have seen H.264 video compression standard. But when we started we had Windows Media, Real Media format, MPEG-2, MPEG-4 and H.264. So another challenge was to be able to tell the content providers that they could provide the media in any format. Integrating all these formats itself was a big challenge.

What are the latest trends in this field? Can smart TVs integrate this entire product into them?

Technically, smart TVs can do so. But when we do that, we have to balance out the benefits in order to make successful business out of it.

The fact is that TV margins are already very thin. So if we add another component to it, we have to balance the economics of it based on how much value it gives. On one hand, the value addition might attract more buyers to the device. On the other hand, there is the risk that consumers may return the product because one component doesn't work. This risk increases with TV complexity. So if you have bought a 35,000-rupees TV and its one component worth a few thousand rupees is not working, you will definitely return the product. In an industry that survives on such low margins, the product returns might break the company.

However, you could see this function coming in premier TVs. The cost of these TVs includes a premium for the advanced technology of the device integrated into the TV.

What's your take on Apple TV and Google TV?

Apple TV has not affected us because it is a closed product. To get content, you have to sign into the iTunes Store. In our case, you have the choice to get the content from us or from any other content provider. You could even open a content store yourself.

In some ways, Apple TV did a lot of good and bad to the industry. For non-techie consumers, it sold the concept that they just take a device, attach it to the Internet and it will work. That was the good part. The bad part was that it didn't seem like a mainstream product for Apple. Steve Jobs even called it a hobby. So people didn't attach a sense of failure to that product. But its net result was positive as it created a lot of consumer awareness about it.

Google TV is very similar. A lot of people from the outside look at Google TV as a failure. But, technically, they did a very good job. In the TV industry, what we have learnt is that it is all about user experience. Microsoft has been trying this since 1998 after it bought a company called webTV. If you have decided in your mind to look at the TV as a browsing device, it's not going to work in this

generation. TV users do not like a lot of interactivity. The use of a keyboard with TV is very alien to most people. TV watching is by far still a passive experience.

When Google TV came out, they focused on the integrated search experience—you could search for content from the Internet or the cable operators. It was technically impressive but there was a huge change in the user experience. I personally don't believe that any change of that magnitude will happen very quickly.

What is VuNow?

We are not just a box company. Even though there is a significant amount of IP in the box, it also talks to a head-end on the Internet. So this whole cloud service that manages the content completely comes from us. The whole system is called the VuNow Internet TV platform. That is what our customers take from us. They go to end users with different brand names for the box.

What challenges did you face in bringing this product to the market?

I will talk about the external factors first. It's a very embryonic market, not a mature market. Entering an embryonic market means you have to face a lot of challenges. For one, we don't go to the consumer directly, so we don't get the consumer feedback directly. So the whole loop of getting the product to the customer and then to the consumer, obtaining the feedback, and then iterating through that and building the right stuff into the product itself is a challenge.

The market is in a developing stage, and people really don't know in which direction it might go. After we rolled out the first version of the product, Apple TV was announced. Apple did a lot of marketing for the entire eco-system, but they had a different vision. They are good at building this into the minds of the consumer right. People might tend to follow them, but later they realise that this is not right and then they come back to us.

The second challenge is internal. Engineers are used to working in a certain way. When you build a whole product all the way from very serious things like deep design and architecture, it requires thinking over the long term. Also the productisation does not come naturally to a lot of engineers.

Is India ready for this type of product?

India is one of the few countries where we have lesser penetration. The reason is very simple: This is a product that delivers video over the Internet. So if I want to deliver high-quality video over the Internet, I am talking about having at least a 2Mbps connection. That's why it's US and Western Europe where all the action is happening right now. India is developing, so we'll be looking here at a larger scale soon. ●